

Week 2 – Python for Data Analysis

Complete Practice Set Solutions

The uploaded practice set contains five Python/Pandas tasks: loading a CSV and displaying basic information; handling missing values and duplicates; grouping by category to calculate total revenue; sorting by multiple columns; and creating a correlation matrix for numerical columns.

1. Load a CSV File and Display Basic Information

Explanation: `pd.read_csv()` loads the CSV into a Pandas DataFrame. `head()` previews the records, `shape` gives the number of rows and columns, `columns.tolist()` displays column names, `info()` shows data types and missing-value information, and `describe()` provides summary statistics.

```
import pandas as pd

# Load the CSV file
df = pd.read_csv("global_superstore_2016.csv")

# Display first 5 rows
print(df.head())

# Display dataset shape
print("Rows and Columns:", df.shape)

# Display column names
print("Columns:", df.columns.tolist())

# Display data types and non-null counts
df.info()

# Display summary statistics
print(df.describe(include="all"))
```

2. Handle Missing Values and Duplicates

Explanation: `isnull().sum()` identifies missing values column by column. `duplicated().sum()` counts duplicate records. `drop_duplicates()` removes repeated rows. Numeric missing values can be replaced with the median, while missing categorical/text values can be replaced with a label such as Unknown.

```
# Check missing values
print(df.isnull().sum())

# Check duplicate rows
print("Duplicate rows:", df.duplicated().sum())

# Remove duplicate rows
df = df.drop_duplicates().copy()

# Example: fill numeric missing values with the median
numeric_cols = df.select_dtypes(include="number").columns
df[numeric_cols] = df[numeric_cols].fillna(df[numeric_cols].median())

# Example: fill text/category missing values with "Unknown"
text_cols = df.select_dtypes(include="object").columns
df[text_cols] = df[text_cols].fillna("Unknown")

# Verify again
print("Missing values after cleaning:")
print(df.isnull().sum())
```

```
print("Duplicate rows after cleaning:", df.duplicated().sum())
```

3. Group Data by Category and Find Total Revenue

Explanation: `groupby()` creates groups based on Category. `sum()` adds the Sales values within each category. The result is then sorted so the categories with higher total revenue appear first. If your dataset uses a column named Revenue instead of Sales, replace Sales with Revenue.

```
# Calculate total revenue for each category
category_revenue = (
    df.groupby("Category", as_index=False)["Sales"]
      .sum()
      .rename(columns={"Sales": "Total Revenue"})
)

# Sort from highest to lowest revenue
category_revenue = category_revenue.sort_values(
    "Total Revenue", ascending=False
)

print(category_revenue)
```

4. Sort Data by Multiple Columns

Explanation: `sort_values()` can sort using more than one column. Here Category is sorted alphabetically (`ascending=True`), while Sales is sorted from highest to lowest within each category (`ascending=False`).

```
# Sort first by Category and then by Sales
sorted_df = df.sort_values(
    by=["Category", "Sales"],
    ascending=[True, False]
)

print(sorted_df.head(20))
```

5. Create a Correlation Matrix for Numerical Columns

Explanation: `select_dtypes(include='number')` keeps numerical columns only. `corr()` calculates pairwise correlation coefficients. A value close to +1 indicates a strong positive linear relationship, a value close to -1 indicates a strong negative relationship, and a value near 0 indicates little linear relationship.

```
# Select numerical columns
numeric_df = df.select_dtypes(include="number")

# Calculate correlation matrix
correlation_matrix = numeric_df.corr()

print(correlation_matrix)

# Optional: visualize the correlation matrix
import matplotlib.pyplot as plt

plt.figure(figsize=(10, 7))
plt.imshow(correlation_matrix, cmap="coolwarm", interpolation="nearest")
plt.colorbar()
plt.xticks(range(len(correlation_matrix.columns)),
            correlation_matrix.columns, rotation=45, ha="right")
plt.yticks(range(len(correlation_matrix.columns)),
            correlation_matrix.columns)
plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()
```

Complete Combined Script

```
import pandas as pd
import matplotlib.pyplot as plt

# 1. Load data
df = pd.read_csv("global_superstore_2016.csv")
print(df.head())
print("Shape:", df.shape)
print("Columns:", df.columns.tolist())
df.info()

# 2. Clean missing values and duplicates
print("Missing values before cleaning:")
print(df.isnull().sum())
print("Duplicates before cleaning:", df.duplicated().sum())

df = df.drop_duplicates().copy()

numeric_cols = df.select_dtypes(include="number").columns
df[numeric_cols] = df[numeric_cols].fillna(df[numeric_cols].median())

text_cols = df.select_dtypes(include="object").columns
df[text_cols] = df[text_cols].fillna("Unknown")

print("Missing values after cleaning:")
print(df.isnull().sum())
print("Duplicates after cleaning:", df.duplicated().sum())

# 3. Total revenue by category
category_revenue = (
    df.groupby("Category", as_index=False)["Sales"]
      .sum()
      .rename(columns={"Sales": "Total Revenue"})
      .sort_values("Total Revenue", ascending=False)
)
print(category_revenue)

# 4. Sort by multiple columns
sorted_df = df.sort_values(
    by=["Category", "Sales"],
    ascending=[True, False]
)
print(sorted_df.head(20))

# 5. Correlation matrix
numeric_df = df.select_dtypes(include="number")
correlation_matrix = numeric_df.corr()
print(correlation_matrix)

plt.figure(figsize=(10, 7))
plt.imshow(correlation_matrix, cmap="coolwarm", interpolation="nearest")
plt.colorbar()
plt.xticks(range(len(correlation_matrix.columns)),
            correlation_matrix.columns, rotation=45, ha="right")
plt.yticks(range(len(correlation_matrix.columns)),
            correlation_matrix.columns)
plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()
```

Note: The uploaded Week 2 practice set specifies the tasks but does not provide a dataset schema in the PDF. The examples above therefore assume a Global Superstore-style file with a Category column and a Sales column. If your actual CSV uses different column names, replace those names in the relevant code.